Design of Lead-Lag Compensation via Root-Locus
– Lag Compensation

(通过根轨迹的超前-滞后补偿设计 – 滞后补偿器)

Lag Compensation

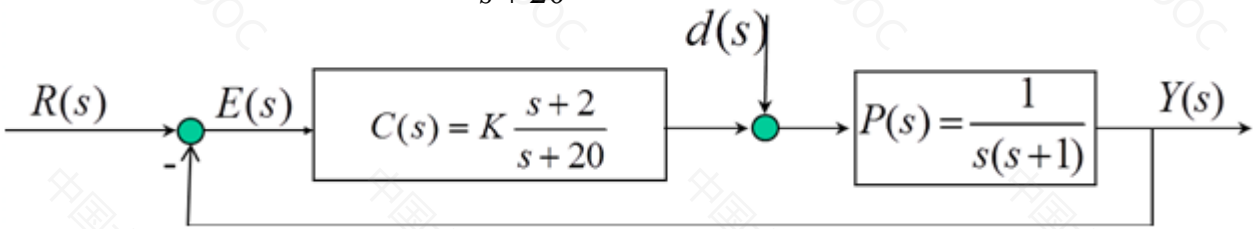
- **Purpose (目的):**
  - Essentially, a lag-compensator adds a pole very close to the origin (本质上一个滞后补偿器增加了非常接近于原点的极点)

    like the integral control adding a pole at origin to reduce steady-state error. (就像积分控制增加了1个在原点的极点从而降低了稳态误差)
  - Leave the original root locus undisturbed (原始根轨迹不受影响)

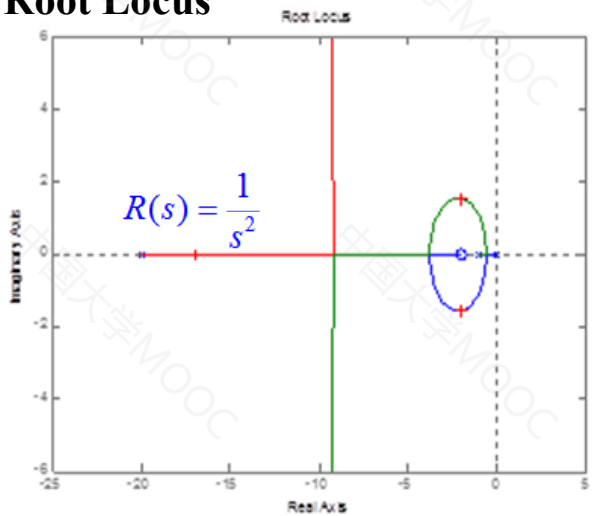
    The transient performance achieved with existing compensation will not be altered. (采用现有补偿器的系统瞬态性能不会被改变)
  - Therefore, place a zero very close to the pole to cancel its effect (similar to the pole/zero cancellation) (放置一个接近于极点的零点以抵消其影响(类似于零极点相消)).
  - In this way, the newly created branch of root locus may not affect other branches seriously to change the satisfactory dynamic response achieved by previously designed lead compensators (通过这种方法，新的根轨迹分支不可能严重影响其他的分支，从而由之前设计的超前补偿器获得满意的动态性能)

Lag Compensation

Ex. Lead Compensation $K\dfrac{s+2}{s+20}$ with $K=55$



➢**Root Locus**



➢**CL poles (闭环极点)** $p_{CL1,2}=-2\pm1.55j,$ $p_{CL3}=-17$

➢**CLTF (闭环传递函数)**

$$\frac{E(s)}{R(s)}=\frac{s(s+1)(s+20)}{s(s+1)(s+20)+K(s+2)}$$

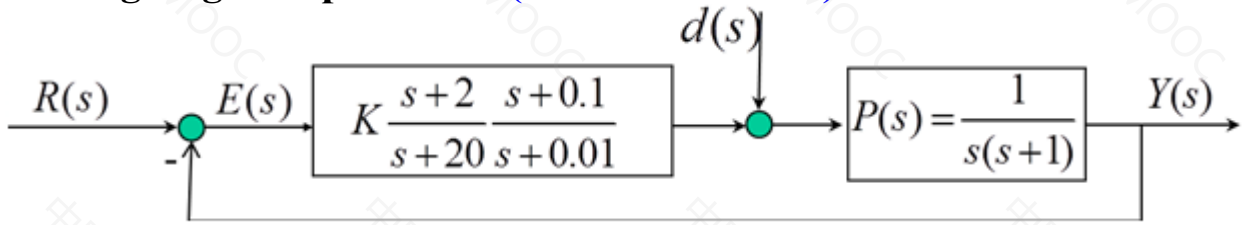
➢**Steady-state Error for a unit Rump Input (单位斜坡输入的稳态误差)** ($R(s)=\dfrac{1}{s^2}$)

$$\lim_{s\to0}e(t)=\lim_{s\to0}sE(s)=\lim_{s\to0}\frac{(s+1)(s+20)}{s(s+1)(s+20)+K(s+2)}$$

$$=\frac{20}{2K}=\frac{2}{11}$$

Lag Compensation

➢**Adding Lag Compensation (增加滞后补偿器)**



➢**CLTF (闭环传递函数)**

$$T(s)=\frac{K\dfrac{(s+2)}{(s+20)}\dfrac{(s+0.1)}{(s+0.01)}\dfrac{1}{s(s+1)}}{1+K\dfrac{(s+2)}{(s+20)}\dfrac{(s+0.1)}{(s+0.01)}\dfrac{1}{s(s+1)}}=\frac{K(s+2)(s+0.1)}{s(s+1)(s+20)(s+0.01)+K(s+2)(s+0.1)}$$

➢**Velocity Constant (速度常数)**

$$\lim_{s\to 0} sL(s)=\lim_{s\to 0} sK\frac{(s+2)(s+0.1)}{(s+20)(s+0.01)}\frac{1}{s(s+1)}=K$$

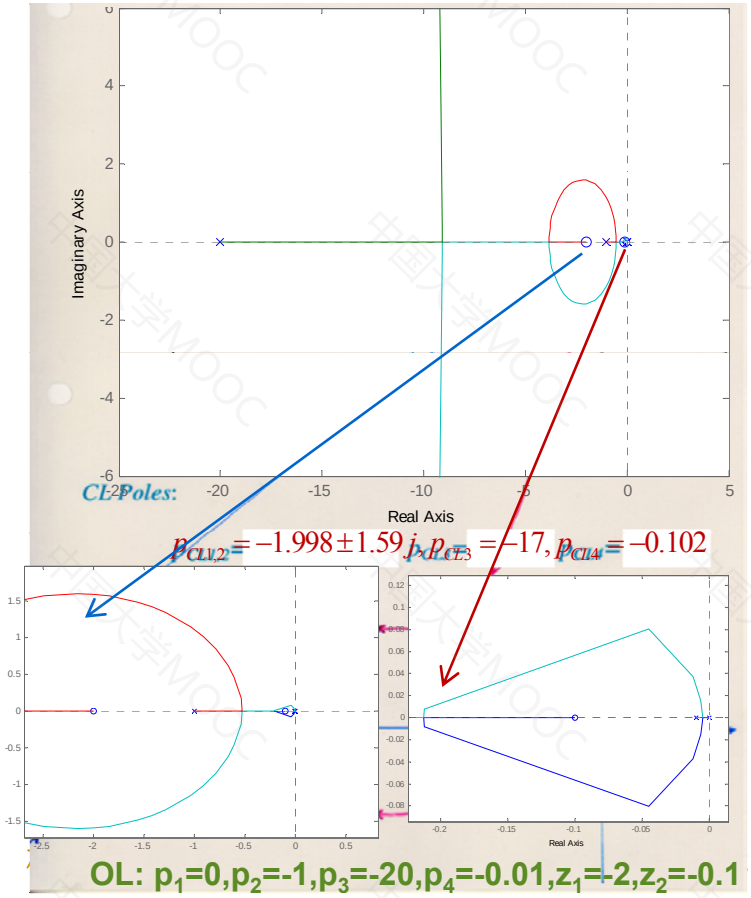
➢**Steady-state Error for a Unit Ramp Input (单位斜坡输入的稳态误差) ($R(s)=\frac{1}{s^2}$)**

$$\lim_{t\to\infty} e(t)=\lim_{s\to 0} sE(s)=\lim_{s\to 0}\frac{(s+1)(s+20)(s+0.01)}{s(s+1)(s+20)(s+0.01)+K(s+2)(s+0.1)}=\frac{0.2}{0.2K}=\frac{1}{55}$$

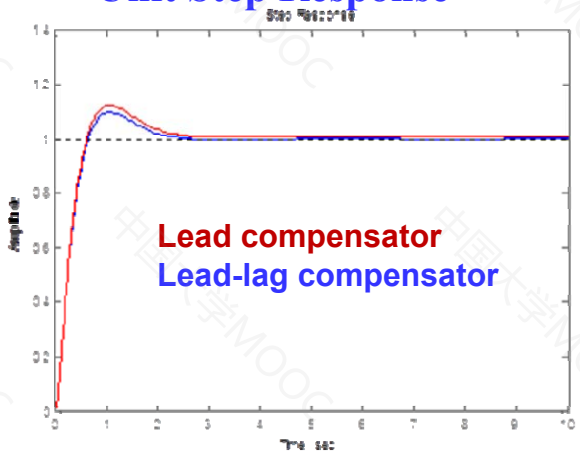
Notice that the reduction ratio is 0.1/0.01=10! (注意降低了10倍)

Lag Compensation

Root Locus



Unit Step Response



- The added lag compensation results in a very slow CL pole near the zero of the lag compensation (p_{CL4}=-0.102) **(增加的滞后控制器引起系统在滞后控制器的零点附近有非常慢的闭环极点p_{CL4}=-0.102)**
- Due to the near pole/zero cancellation effect of the lag compensation, the added slow CL pole does not affect the transient response much **(由于滞后控制器非常接近的零极点相消了，增加的慢闭环极点没有过多的影响瞬态响应)**

Guidelines for Designing Lag Compensator

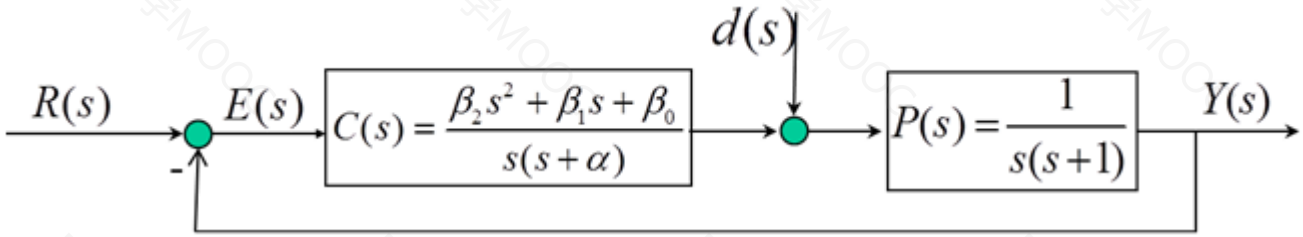
- The pole and zero should be placed far apart to get enough reduction in steady-state error, but also close together so that they do not affect other root loci much (极点和零点应当配置得足够远以便降低稳态误差，但两者又相当接近以便不过多的影响根轨迹).
- Adding a lag compensation will add one additional CL pole to the resulting system, which is forced to stay close to the pole-zero pair of the lag compensator and thus is near origin (增加1个滞后控制器将会对最终系统增加一个额外的闭环极点，必须使滞后控制器具有相当接近的零极点且它们离原点很近).
- As a result, the closed-loop response may become sluggish due to the slow response associated this added CL pole. To avoid this, the pole-zero pair should be placed to the left as far as possible while without affecting other root loci too much (由于增加了慢响应的闭环极点，可能会使闭环响应变慢。为避免这种情况，零极点对在不较大影响根轨迹的情况下应当尽可能靠左).

The additional CL pole due to the added lag compensator is somewhere in between the pole and zero of the lag compensator. If the pole and the zero of the lag compensator are far apart, this additional slow pole may not be near the zero of the lag compensator (one of the CL zeros), and thus its effect may be significant since we do not have the pole/zero cancellation phenomenon any more (由于增加的闭环极点处在所加入的滞后控制器的零点和极点之间的某个地方，这个额外的慢极点可能不在滞后补偿器的零点附近(其中1个闭环零点)，由于没有零极点相消现象,其影响可能会比较显著).

Procedure for Designing Lag Compensator

- Sketch the root locus of the uncompensated system (the system may already be compensated by the lead compensator) (画出未补偿系统的根轨迹(系统可能已经由超前控制器补偿过)).
- Determine the desired roots from the plot. Find the value of K (从图中确定期望根轨迹，找出K值).
- Calculate the steady-state performance. If the steady-state error using current K is n times larger than the desired one, design a lag compensator that is roughly in the form of $\frac{s+n\alpha}{s+\alpha}$. The choice of α may need iterations (计算稳态性能，如果采用当前K的稳态误差比期望的大n倍，则设计滞后控制器的大致形式为 $\frac{s+n\alpha}{s+\alpha}$，其中$\alpha$ 可能需要反复尝试).
  - ✓ Multi-stage lead compensation design can meet design specifications that may not be achieved by a single lead compensator (可能单一超前补偿器不能满足设计要求，可用多段超前补偿器来满足).
  - ✓ The design of each compensator can be carried out independently, i.e., the α and β of each compensator may be different (每个补偿器独立设计，也就是每个补偿器的α和β是不同的).
  - ✓ The use of more than one lag compensator should be avoided (尽量避免采用多于1个的滞后补偿器).
  - ✓ A better design to improve the steady-state tracking performance may be to insert integrators in the controller to raise the system type (提高稳态跟踪性能的更好方法可能是在控制器中加入积分项来提高系统类型)

Ex. Alternative Design to Improve Steady-state Performance



$$T(s)=\frac{Y(s)}{R(s)}=\frac{C(s)P(s)}{1+C(s)P(s)} \qquad T_d(s)=\frac{Y(s)}{d(s)}=\frac{P(s)}{1+C(s)P(s)} \qquad C(s)=\frac{\beta_2 s^2+\beta_1 s+\beta_0}{s(s+\alpha)}=\frac{\beta_2(s-z_{c1})(s-z_{c2})}{s(s+\alpha)}$$

$\Rightarrow$

$$T(s)=\frac{\beta_2 s^2+\beta_1 s+\beta_0}{s(s+\alpha)s(s+1)+\beta_2 s^2+\beta_1 s+\beta_0}=\frac{\beta_2(s-z_{c1})(s-z_{c2})}{s^4+(1+\alpha)s^3+(\alpha+\beta_2)s^2+\beta_1 s+\beta_0}$$

$$T_d(s)=\frac{s(s+\alpha)}{s(s+\alpha)s(s+1)+\beta_2 s^2+\beta_1 s+\beta_0}=\frac{s(s+\alpha)}{s^4+(1+\alpha)s^3+(\alpha+\beta_2)s^2+\beta_1 s+\beta_0}$$

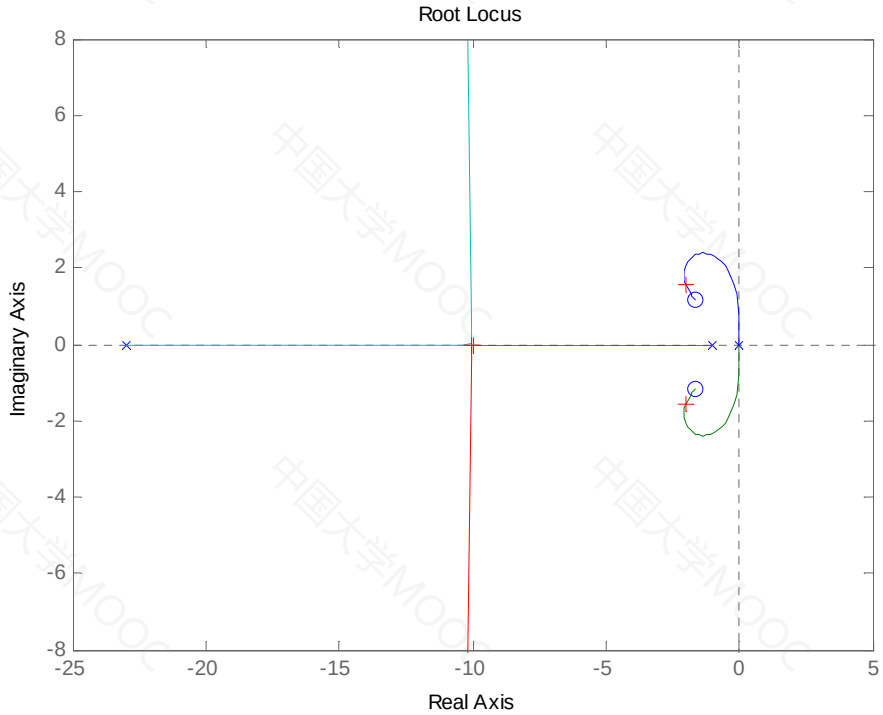
<u>Desired CLCP:</u> $p_{CL1,2}=-2\pm 1.55j,\ p_{CL3,4}=-10$

$$(s-p_{CL1})(s-p_{CL2})(s-p_{CL3})(s-p_{CL4})=\left[(s+2)^2+1.55^2\right](s+10)^2$$

$$=s^4+24s^3+186s^2+528s+640$$

Ex. Alternative Design to Improve Steady-state performance

$$C(s)=\frac{163s^2+528s+640}{s(s+23)}$$



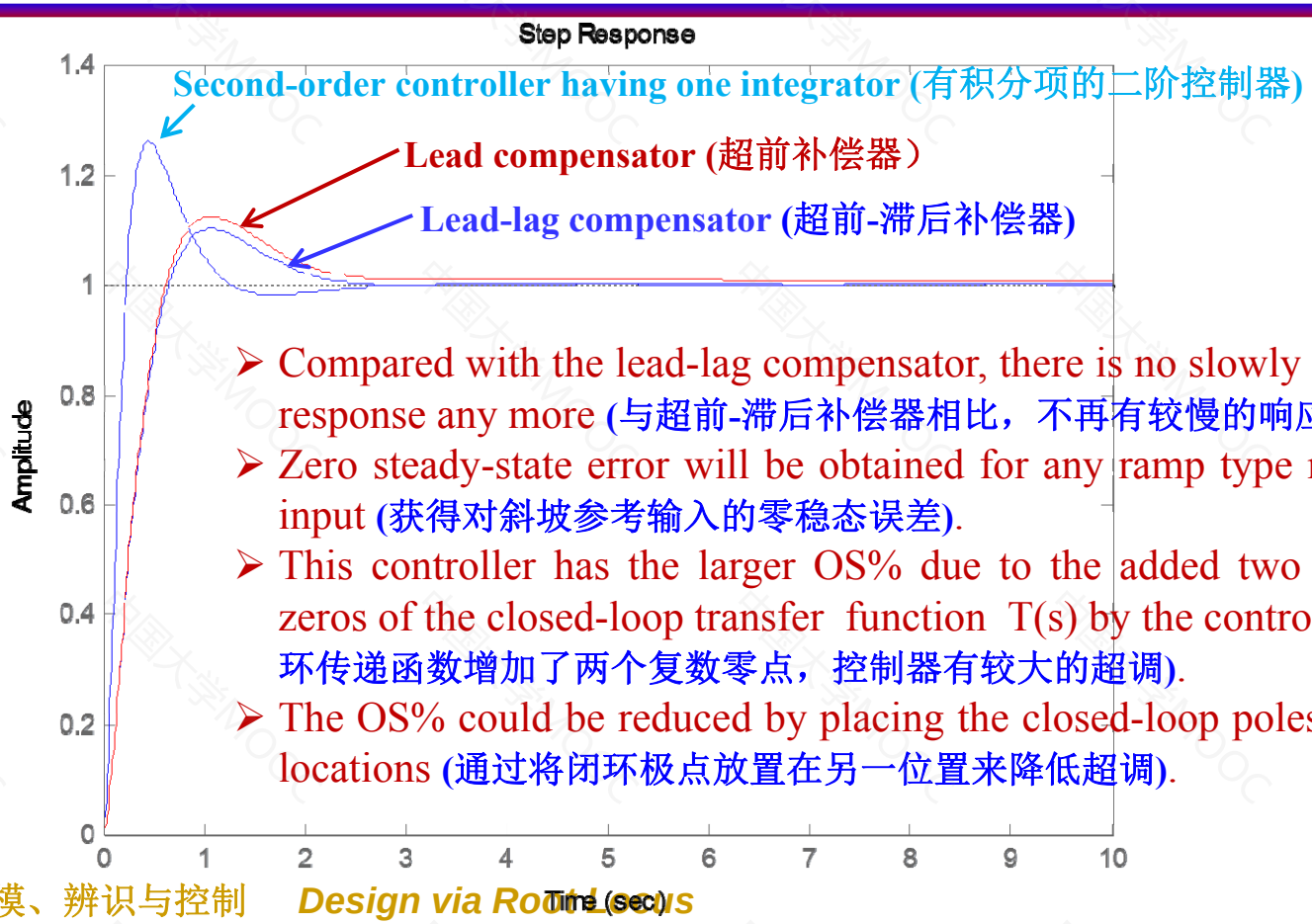
$$z_{c1,2}=-1.616\pm1.143j$$

$$p_{1,2}=0,\ p_3=-23,\ p_4=-1$$

$$p_{CL1,2}=-2\pm1.55j,$$

$$p_{CL3,4}=-10$$

Ex. Alternative Design to Improve Steady-state Performance



- Compared with the lead-lag compensator, there is no slowly decaying response any more (与超前-滞后补偿器相比，不再有较慢的响应).
- Zero steady-state error will be obtained for any ramp type reference input (获得对斜坡参考输入的零稳态误差).
- This controller has the larger OS% due to the added two complex zeros of the closed-loop transfer function T(s) by the controller (因闭环传递函数增加了两个复数零点，控制器有较大的超调).
- The OS% could be reduced by placing the closed-loop poles at other locations (通过将闭环极点放置在另一位置来降低超调).

Ex. Alternative Design to improve steady-state performance

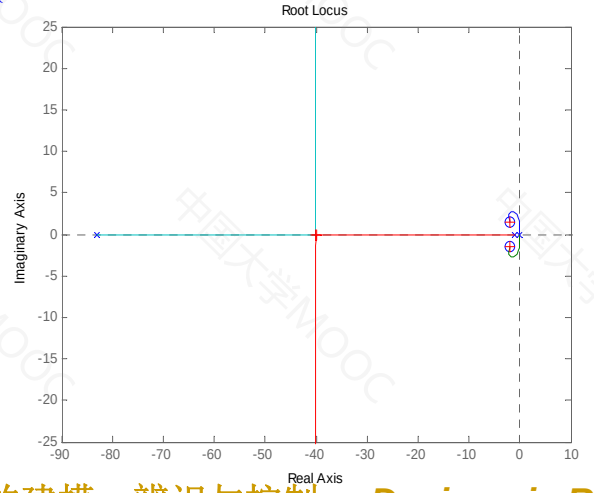
Desired CLCP: $p_{CL1,2} = -2 \pm 1.55j, p_{CL3,4} = -40$

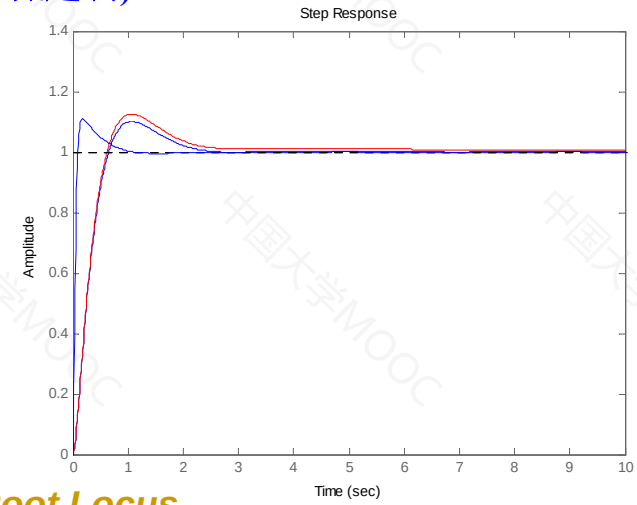
$$(s - p_{CL1})(s - p_{CL2})(s - p_{CL3})(s - p_{CL4}) = \left[(s+2)^2 + 1.55^2\right](s+40)^2$$

$$= s^4 + 84s^3 + 1927s^2 + 6912s + 10244$$

$$C(s) = \frac{1843s^2 + 6912s + 10244}{s(s+83)} = \frac{1843(s+\ldots)(s+\ldots)}{s(s+83)}$$

Overshoot can be reduced by placing CL poles at other locations!
(通过配置闭环极点在其他位置上能降低超调)





Summary

- **Design of Lag Compensation** (滞后补偿器设计)
  - ✓ Guidelines for Designing Lag Compensator (设计滞后补偿器的原则)
  - ✓ Procedure for Designing Lag Compensator (滞后补偿器设计步骤)
  - ✓ Alternative Design to Improve Steady-state Performance (提高稳态性能的备选设计)